Nuclear Physics Lab instructions

March 7, 2024

We shall discuss mechanisms of radiation attenuation by matter, as well as some statistical aspects of the measurements. In particular we shall discuss why the number of particles detected during a fixed time interval obeys the Poisson distribution, and how this distribution can be distinguished from the Gaussian one.

Note: For the first reading, you may focus on the highlighted paragraphs.

Contents

1 Introduction

We shall focus on radiation emitted by nuclei, specifically on three distinct types of radiation. Below is the excerpt from the Nobel Prize lecture, Chemistry 1908, by Ernest Rütherford. We would recommend to read the Nobel Prize lecture for the fascinating history of radioactivity discovery.
"..these experiments show that the uranium radiation is complex and that there are present at least two distinct types of radiation – one that is very readily absorbed, which will be termed for convenience the α-radiation, and the other of a more penetrative character, which will be termed the β-radiation." When other radioactive substances were discovered, it was seen that the types of radiation present were analogous to the β– and α-rays of uranium and when a still more penetrating type of radiation from radium was discovered by Villard, the term γ-rays was applied to them. The names thus given soon came into general use as a convenient nomenclature for the three distinct types of radiation emitted from uranium, radium, thorium, and actinium. On account of their insignificant penetrating power, the α-rays were at first considered of little importance and attention was mainly directed to the more penetrating β-rays. With the advent of active preparations of radium, Giesel in 1899 showed that the β-rays from this substance were easily deflected by a magnetic field in the same direction as a stream of cathode rays and consequently appeared to be a stream of projected particles carrying a negative charge. The proof of the identity of the β-particles with the electrons constituting the cathode rays was completed in 1900 by Becquerel, who showed that the β-particles from radium had about the same small mass as the electrons and were projected at a speed comparable with the velocity of light."

2 Background

2.1 α-particles

An α-particle consists of two protons and two neutrons and is a nucleus of helium He. α-particles are emitted spontaneously by unstable heavy nuclei. This emission is rooted in the principles of quantum mechanics: Within an atomic nucleus, protons and neutrons are held by the powerful strong nuclear force, which counteracts the electromagnetic repulsion between positively charged protons. Quantum tunneling, a manifestation of the wave nature of particles at the quantum level, enables an α particle to surmount the energy barrier and pass through it despite lacking sufficient classical energy to pass above it. This emission is represented by the following equation:

$$ {}^{A}_{Z}X \rightarrow^{A-4}_{Z-2} Y +^{4}_{2} \alpha \tag{1}$$

where X and Y are the initial and final nuclei, A is the mass number (sum of the numbers of protons and neutrons in the nucleus), Z is the atomic number (number of protons in the nucleus) and α represents the emitted α-particle.

α-particles emitted from a particular species of nuclei are mono-energetic. The energy of the most α's falls in the range between 3 and 8 MeV. By comparison, the typical energies of the chemical reactions are between 5 to 30 eV per molecule.
Preparatory question: what is typical speed of α-particles compared to that of valence electrons? Is it a good approximation to consider electrons as static when discussing α-particle–electron collisions?

2.2 β–particles

Beta particles ($\beta^{\pm}$) are electrons (β^{-}) or positrons (β^{+}) that are emitted from nuclei when a proton or a neutron decay by the weak force, emitting (β^{+}) or (β^{-}) referred to as β-plus and β-minus emission. In a β-minus emission, a neutron (n) becomes a proton (p) emitting an electron, e, and an antineutrino $\bar{\nu}$. The corresponding reaction is:

$$ {}^{A}_{Z}X \rightarrow^{A}_{Z+1} Y + \beta^{-} + \bar{\nu} \tag{2}$$

Unlike α-particles that are emitted with well-defined energy from nuclei, β-particles are emitted with a range of energies between 0, and the maximum energy specific to the given radioactive isotope. The rest of the energy is taken by antineutrino. The maximal energy of the emitted β-particles is a characteristic signature of the emitting isotope. An example of intensity as a function of the energy for β-decay is shown in Fig. 1.

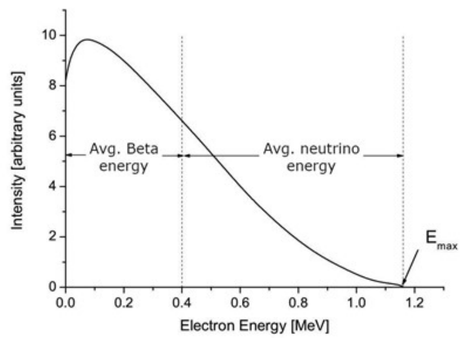

Figure 1: Intensity of β-particle emission vs β-particle energy. The arrow indicates the maximum energy E_{max} of β-particles. The left dashed line represents the average energy of the β-particles, while the difference between the maximum energy and the average energy corresponds to the average energy of neutrinos.

2.3 γ-particles

A γ-particle, an energetic photon emitted by excited nuclei in their transition to lower-lying nuclear levels. It is uncharged, unlike α or β-particles. Since nuclear states have well-defined energies, the energies of γ-particles emitted in state-to-state transitions are also well-defined.

3 Penetration range

In the lab you will measure the thickness of the material which the radiation consisting of charged particle can penetrate before losing its energy due to scattering and absorption processes. This distance is commonly referred to as penetration range. There is some ambiguity in defining penetration range. It is typically defined as either Half-Value Layer (HVL) or Tenth-Value Layer (TVL): The HVL (TVL) is the thickness of a material that reduces the intensity of the radiation to half (tenth) of its original value. We shall adopt the TVL definition. A natural question to ask is why not to fit the radiation intensity to exponent and use the power as the measure of the range. You will see that your experimental data not always can be fit by a single exponent.

In this section we will explain how nuclear radiation interacts with material between the source and the detector. Charged particles primarily interact with atoms in the medium through Coulomb forces. They can excite or ionize atoms, resulting in a decrease in the particles' energy and consequently slowing them down. We shall discuss the particle scattering using the Rütherford's differential cross-section $d\sigma/d\Omega$ formula:

$$\frac{d\sigma}{d\Omega} = \left(\frac{Z_1 Z_2 e^2}{8\pi\varepsilon_0 \mu v^2 \sin^2\Theta/2}\right)^2 \tag{3}$$

$d\sigma/d\Omega$ depends inversely on the square of the reduced mass $\mu = m_1 m_2/(m_1 + m_2)$ of the interacting particles. In Eq. 3 Z_1 and Z_2 are the dimensionless charges of the incident and target particles, e is the elementary charge, ε_0 is the vacuum permittivity, v is the relative velocity between the particles, and θ is the scattering angle *in the center of mass frame*. The same formula can be expressed through the energy E_c of the colliding particles in that frame:

$$\frac{d\sigma}{d\Omega} = a_0^2 \left(\frac{Z_1 Z_2 R_y}{2E_c \sin^2\Theta/2}\right)^2, \tag{4}$$

where $a_0 = 4\pi\varepsilon_0\hbar^2/(m_e e^2)$ is the Bohr radius, and $R_y = (m_e e^4)/(8\varepsilon_0 h^3 c) \approx 13.6\,\text{eV}$ is the Rydberg constant; $a_0^2 \approx 2.8 \cdot 10^{-21} m^2$ sets the cross-section scale.

Historically, he Rütherford formula was obtained by considering Coulomb scattering classically. Quantum treatment in the first Born approximation gives the same result. This approximation assumes that the scattering is weak, so the amplitude of the scattered wave is small compared that of the incident one. We shall analyze α and β-scattering on the basis of this formula, although it assumes both particles to be free, and thus ignores material properties besides very basic ones: electron density and atomic number. Both α and β-particles undergo scattering interactions with both electrons and nuclei. However, their scattering is very different due to almost 4 orders of magnitude difference in masses ($m_\alpha \approx 7.3 \times 10^3 m_e$).

3.1 α-particles range

α-particles are much heavier than electrons, and this determines peculiarities of their scattering.

1. α-particles scatter mostly on electrons. This is because their reduced mass with electrons is small, $\mu_{\alpha-e} \approx m_e$, therefore their scattering cross-section on electrons is much larger than that on nuclei, despite nuclei having larger charge eZ_n. The ratio of the scattering cross-sections of an α-particle on an electron and on a nucleus, from Eq. 3, is $(m_\alpha / Z_n \cdot m_e)^2 \approx 4.5 \cdot 10^4$.

2. Since the mass of an α-particle is much larger than that of an electron, after the scattering on an electron the α-particle is almost not deflected and continue to move in the original direction. Indeed, the conserved center of mass momentum is predominantly carried by the alpha particles. Consequently, the deflection of an α-particle after a scattering on an electron is minimal.

3. Scattering on nuclei does occur, and it is this scattering that led to the original detection of α-particles by Rütherford. However, due to its much smaller cross-section, it is not the scattering that dominates attenuation of an α-particle beam .

4. Most importantly, scattering of α-particles is strong compared to β-particles. This is best seen by looking at Eq. 3 and observing that, for the came energy, the ratio of velocities is: $v_\alpha / v_\beta = \sqrt{m_\beta / m_\alpha}$.

   Problem: Consider a 1 MeV α-particle and an electron at rest. Calculate the kinetic energy of their relative motion E_r in the center of mass frame; E_r is carried mostly by the electron. How does E_r compare with the typical binding energy of an electron in, say, Aluminum ($Z = 13$). To what extent is it justified to consider electrons to be at rest?

Answer: $E_r = E_\alpha m_e/(m_e + m_\alpha) \approx E_\alpha m_e/m_\alpha \approx 136\,\mathrm{eV} \approx 10\,\mathrm{Ry}$. So, considering electrons to be at rest can be justified only for the top occupied levels, and is not justified at all for the deepest ones with energy $\sim Z^2$Ry. But ignoring bounding of deep-lying electrons to the nuclei also is a crude approximation. Still, Rütherford formula is good for estimates.

5. The α-particles range can be approximated as [1]

$$R = 0.32(E_\alpha/\mathrm{Mev}))^{3/2}\times\mathrm{cm} \quad \text{or} \quad R = 0.4(E_\alpha/\mathrm{Mev}))^{3/2}\times\mathrm{mg/cm}^2 \quad (5)$$

Note that R is customary expressed both as length and as areal density, which can be confusing.

3.2 β-particles range

1. The reduced effective mass for electron-electron scattering $\mu_{e-e} = m_e/2$, only twice smaller than that for electron-nucleus one: $\mu_{e-n} = m_e m_n/(m_e + m_n) \approx m_e$. Due to larger nuclei charge eZ, β-particles scatter more on nuclei than on electrons. But not much more.

   Problem: Estimate the ratio of the scattering rates on electrons and on nuclei. Take into account that each nucleus with number Z is surrounded by Z electrons, so the scattering rate is proportional to Z for electrons and Z^2 for nuclei.

2. For the $\beta - e$ scattering, the energy in the center of mass frame is just about half of that in the rest frame. So, the cross-section for β-particles scattering, according to Eq. 4, is rather small compared to that of α-particles with the same energy. We can see it also from Eq. 3 by observing that the speed v_β of β-particles with typical energy of 1 MeV is large.

3. Despite nuclei being effective in β-particle scattering, the particles lose energy through Bremsstrahlung (braking radiation), and this energy loss is small, so the scattering is quasi-elastic.

4. As β-particles possess the same rest mass as electrons, unlike α-particles, they are deflected by scattering much stronger, although forward scattering still dominates. While radiation progresses through the media, it becomes less energetic, and its scattering cross-section increases. This leads to non-exponential decay of the radiation intensity. In addition, secondary electrons and photons can be detected: Energetic

β-particles can dislodge electrons from the media. There is also photon and electron emission when electrons in the media, excited by the radiation, recombine.

5. Since β-particles lose energy mostly by scattering on electrons, their range in different materials is expected to scale inversely with the electron density. Therefore, the material-specific penetration length $R_{(L)}$ in units of length, say (cm), is customarily multiplied by material density ρ, in (g/cm^3):

$$R = R_{(L)} \cdot \rho. \tag{6}$$

   The resultant R in (g/cm^2) is almost material-independent. The range, both in (cm) and in (g/cm^2) is customarily denoted by the same letter R. It would be more accurately to express the range for a material x through the range in the well-studied material, typically Al (Al is light, so the range in cm in it is long and easy to measure) as:

$$R_x = R_{Al} \frac{(Z/A)_{Al}}{(Z/A)_x} \tag{7}$$

   However, since most of the stable elements have almost equal number of protons and neutrons in their nuclei, $Z/A \approx 1/2$, the correction of (7) is small.

6. Energy dependence of the penetration length for Aluminum can be described by the empiric Flammersfeld's formula [2]:

$$R_{Al} \approx 0.11 \left(\sqrt{1 + 22.4(E/\text{MeV})^2} - 1 \right) \times \text{g/cm}^2 \tag{8}$$

   One may apply correction Eq. 7 if more accuracy is needed. There are other approximations for β-particles range.

3.3 γ-ray attenuation

Gamma-particles (γ), which are energetic photons emitted by nuclei, are chargeless. Therefore, their scattering cross-section is smaller compared to that of β-particles. To absorb γ-radiation to one tenth of its original intensity, a thick material is necessary. For instance, approximately 5 cm of iron is required for $E_\gamma = 1\,\text{MeV}$. For this reason, heavy materials, most commonly lead, are used to screen γ and even less-energetic X-ray radiation.

Following processes are mainly responsible for γ-ray attenuation:

1. With similar reservations as for α and β-particles, the scattering of uncharged γ-particles on electrons, known as Compton scattering, can be described using the Klein–Nishina formula:

$$\frac{d\sigma}{d\Omega} = \frac{1}{2} r_e^2 \left(\frac{\lambda}{\lambda'} \right) \left[\frac{\lambda}{\lambda'} + \frac{\lambda'}{\lambda} - \sin^2(\theta) \right], \tag{9}$$

   where λ and λ' are initial and final wavelengths of the photon. Importantly, the cross-section scale in 9 is the square of the classical electron radius $r_e \approx 2.82 \cdot 10^{-15}$ m. Therefore the prefactor in Klein-Nishina's formula is smaller than that of the Rütherford's one by 9 orders of magnitude, and this process is not effective in γ-ray attenuation.

2. Photoelectric effect. Its cross-section is about $10^{-27}\,\mathrm{m}^2$, and rapidly increases with decreasing energy.

3. Electron-positron pair formation: occurs only for energies above 1.02 Mev, and its cross-section is small.

So, the γ-ray attenuation is mostly due to the photo-effect, and is much smaller than that of the β.

3.4 Summary

To summarize, α-particles are predominantly scatter on electrons and continue their motion almost straight along the original path. Their scattering cross-section is large, and therefore their penetration range is small.
β-particles scatter on both electrons and nuclei. Their scattering cross-section is significantly smaller than that of α-particles. This allows them to better penetrate through matter. While α-particles can be stopped by a few centimeters of air, β-particles can travel tens centimeters to a few meters in air before being fully scattered. For instance, α-particles emitted by Uranium-238 with energy 4.36 MeV have range 2.6 cm in air, while beta particles emitted by Carbon-14 with $E_{max} = 0.156$ MeV– approximately 22 cm, and by Phosphorus-32 with $E_{max} = 1.71$ MeV about 6 m.
γ-particles penetrate most, and therefore their penetration is usually characterized by the attenuation coefficient and not by penetration range.
The simple treatment above correctly captures the hierarchy of the attenuation of the different types of radiation, despite ignoring some processes that (i) attenuate radiation and (ii) can lead to secondary radiation. For instance, both α and β-particles can dislodge electrons from the media, and these secondary electrons can be detected. They can also lead to both photon and electron emission when electrons excited by the radiation recombine. While radiation progresses through the media, it becomes less energetic, and therefore its scattering cross-section increases. All this together leads to non-exponential decay of the radiation intensity. Still, for relatively low-energy β-particles as emitted by Thallium-204, the exponential dependence fit the data relatively good.

4 Radiation statistics

Suppose that distinct nuclei independently emit γ-particles or undergo α or β-decay. It is not always the case: think e.g. about A-bomb as an example, where stimulated chain reaction is initiated by spontaneous decay ^{235}U or ^{239}Pu. However, with very weak radiation sources you will be using, stimulated reactions are negligible. So, we expect the observed radiation to obey Poisson statistics, see below.
Let us start with notations: the mean value of n is denoted by overbar: $\bar{n}$. The average over a distribution value as $\langle n\rangle$. $\bar{n} \to \langle n\rangle$ for infinite number of trials.

1. The Poisson distribution expresses the probability $P(n)$ of a given number events n events to occur during a fixed time interval τ, assuming that these events are independent of each other and occur with a known constant mean rate λ.
$$P(n) = \frac{\lambda^n e^{-\lambda}}{n!} \tag{10}$$
A derivation of this formula can be found in the Appendix A.1.

2. The Poisson distribution possesses a property that all its *cumulants* are equal to λ. In particular, the average is $\kappa^{(1)} = \langle n\rangle = \lambda$, the variance is $\kappa^{(2)} = \sigma^2 = \langle (n-\lambda)^2\rangle = \langle n^2\rangle - \lambda^2 = \lambda$, and the skewness $\kappa^{(3)} = \langle (n - \langle n\rangle)^3\rangle = \lambda$.

3. The relationship $\sigma^2 = \lambda$ establishes the connection between the number of observed events and the expected Standard Deviation (STD) of the observations $\sigma = \sqrt{\lambda}$. This is an important distinction from the Gaussian distribution, where the mean and the STD are independent. It facilitates straightforward estimation of the measurement error for values resulting from the Poisson process.

4. Further substantiation of the Poisson process can be attained by examining the third cumulant or skewness $\kappa^{(3)}$. For the Poisson distribution, $\kappa^{(3)} = \langle n\rangle = \lambda$, while for the Gaussian one $\kappa^{(3)} = 0$.

5. You shall utilize the fact that $\sigma = \sqrt{\lambda}$, implying that you anticipate the standard deviation (STD) of a measurement yielding n counts to be $\sqrt{n}$. This is important for estimating the accuracy of your measurements and for including error bars when presenting your data.

In one of the experiments, you will check whether the radiation-generated GM counts are consistent with the Poisson statistics. You will do it by repeating measurements many times, and examining the mean values of the cumulants calculated from these measurements:

$$K_m^{(1)} = \bar{n} = \overline{n}, \quad K_m^{(2)} = \overline{(n-\bar{n})^2} \quad \text{and} \quad K_m^{(3)} = \overline{(n-\bar{n})^3} \tag{11}$$

where n is an element of a set of the measurement results, and the averaging is over the set of m measurements. We reserve notation $K_m^{(i)}$ for cumulants as calculated from the *measured* values averaged over m measurements, to distinguish them from the *expected* $\kappa^{(i)}$ from the Poisson distribution.
For the Poisson process, as the number of measurements m approaches infinity all $K_m^{(i)}$ should converge to the same value λ. Let us estimate the number of measurements required to check that. For this, we shall look at $\sigma_{K_m^{(i)}}/\kappa^{(i)}$, where $\sigma_{K_m^{(i)}}$ is the expectation value of the STD of the i'th cumulant calculated from m measurements.
One can show, see Appendix A.2, that:

$$\sigma_{K_m^{(1)}}/\kappa^{(1)} = 1/\sqrt{\lambda m}, \quad \sigma_{K_m^{(2)}}/\kappa^{(2)} = \sqrt{2/m}, \quad \sigma_{K_m^{(3)}}/\kappa^{(3)} = \sqrt{15\lambda/m} \tag{12}$$

Estimates (12) tell us that:

1. The cumulants converge to the expectation value as $1/\sqrt{m}$. However, achieving the same level of accuracy requires significantly different number of measurements, that increases with the cumulant number.

2. For the first cumulant, $\sigma_{K_m^{(1)}}/\kappa^{(1)} = \delta n/\bar{n} = 1/\sqrt{N}$, as expected. So, in order to get an accurate measurement rate it does not matter how to divide the measurements into intervals. Just the total measurement number N matters, also as expected.

3. For "uncertainty of uncertainty" $\sigma_{K_m^{(2)}}/\kappa^{(2)}$ to be small, one does need to repeat measurements many times.

4. To get a small uncertainty of the skewness, a much larger number of measurements proportional to λ is required.

5 Experimental setup

The setup of the experiment consists of ST-360 pulse counter with GM tube and a stand, as shown in Figure 6. The relevant parameters are in Table 1.

Figure 2: The experimental setup: the GM detector positioned on top of a shelf box, set of Aluminum plates of different thickness, spacers of different thickness, and radioactive sources.

Distance between the shelves	10 mm
Distance between the top shelf and the GM window	12.3 mm
GM window diameter	31.6 mm
Sample shelf thickness	1 mm
GM window density	$2\,\mathrm{mg/cm^2}$

Table 1: Setup parameters

5.1 Geiger Müler counter

In your experiments, you will utilize a Geiger-Müller (GM) counter, based on the Geiger tube. Developed by Geiger and Müller in 1928, this early radiation detector remains widely used due to its simplicity, low cost, and ease of operation. The GM tube contains a noble gas, typically argon. When radiation enters the tube, it ionizes the gas generating free electrons. An ionization event that results in just one or a few free electrons would be challenging to detect. Therefore, the detector is designed to generate an avalanche in response to a few electrons. For this purpose, electrons are accelerated by high voltage between the anode and the cathode of the tube, wherein electrons ionized by incident radiation gain enough kinetic energy to collision-ionize another atom. This process generates avalanches, as illustrated in Fig. 3 and, consequently, current pulses that are detected. A threshold voltage is required for an avalanche to occur. During an avalanche event, additional events cannot be resolved, leading to GM dead time.

Avalanches are sustained, in part, by ultraviolet photon emission resulting from electron recombination with ionized inert gas. This prolongs the dead time. To reduce the inactive period, methane or other quencher molecules are added to the noble gas. These molecules effectively absorb UV photons emitted during the discharge, converting them into heat without generating additional free electrons.

Before entering a GM tube, radiation must pass through a thin window, typically made of mica–a low-density material that allows particles to pass through with little attenuation. It is important to note that, due to their short range, the interaction of alpha particles with a GM window significantly reduces their apparent range. This factor should be taken into account when interpreting experimental results.

GM detectors are less sensitive to gamma radiation, because gamma rays have smaller interaction cross-section with the gas in the tube (and with matter in general) compared to charged α or β-particles. Geiger-Müller tubes are not identical, and require different voltage for optimal performance. When a radioactive sample is positioned beneath the tube and the voltage of the tube is ramped up from zero, the tube does not start firing right away; it needs a minimal voltage for an electron avalanche to start. As the voltage is increased beyond this point, the counting rate increases quickly before stabilizing. The stabilization starts above a voltage commonly referred to as the knee. Further voltage increase above the knee is expected to lead only to moderate increase in the count rate, until the voltage becomes so high that breakdown and spontaneous discharge occur; voltage on GM tube must be kept below the breakdown.

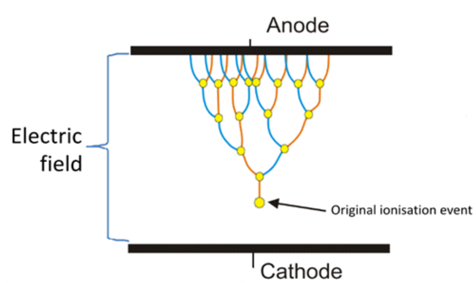

Figure 3: Electron avalanche in GM tube

5.2 Inverse-square law

Below we discuss how radiation flux per unit area would change in the absence of scattering or absorption by medium. As a source is moved farther away from the detector, the detected radiation intensity would decrease proportionally to the solid angle Ω under which the detector window is seen from the source; we consider the source to be a point one. The angle Ω can be expressed through the radius of the detector window r and the distance to the source h, see Fig. 4 as:

$$\Omega = 2\pi(1 - \cos\theta) = 2\pi(1 - 1/\sqrt{(1 + (r/h)^2)}) \qquad (13)$$

For large distances $\Omega \approx 2\pi r^2/h^2$.

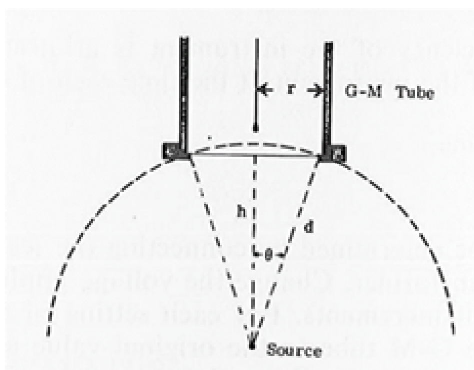

Figure 4: The experimental geometry.

6 Step-by-Step Experiment Procedures

Take a note of the date written on the radioactive sources you are using, and write them in your report. The sources should be placed label down to avoid attenuation by the label. A possible exception – the statistics experiment.

1. Turn on the power switch on the back of the ST-360.
2. Double-click the STX software icon to start the program. You should see the blue control panel appear on your screen, as in Fig. 5.
3. Set High Voltage (HV) to 900V.

4. To start a measurement, press the green “diamond” button. “Preset Time” is the time allotted for each measurement. You can stop the measurement any time, and the “Elapsed time” will be the time that has passed since the start of the measurement.

5. The system is designed to do a set of measurements with a preset time, but I would encourage you to work differently. Since the relative error is $1/\sqrt{n}$, where n is the number of counts, set a relatively long measurement time, and stop the measurements manually when required number of counts is obtained. This will save you time when the count rate is high, and you would be able to use it when it is low. Write down into your Python notebook both the number of counts and the elapsed time. The number of counts will be used also to plot error bars.

6. Aim at 3% error (about 1000 counts) when the rate is high. Be judgemental when it is low: you might need to sacrifice accuracy in order to finish your experiments.

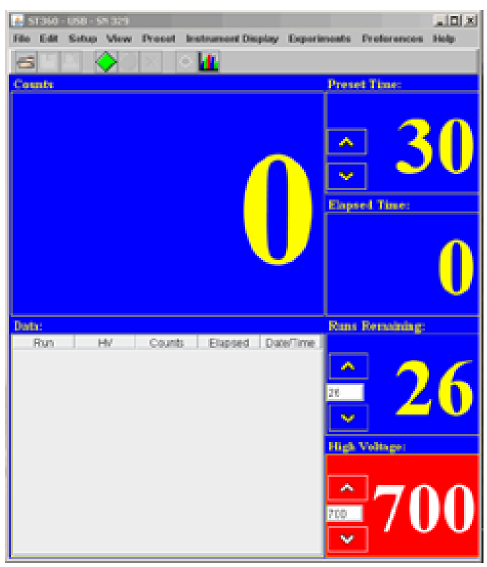

Figure 5: The STX software interface.

6.1 Background radiation

1. Set the measurement time to 100 s.

2. With all sources in the lead cylinder, or at least far away from the GM tube, measure background radiation.

3. Calculate the background radiation rate I_b, and save it for the future. It might need to be subtracted from the future measurements.

6.2 Poisson statistics

Objective: In this experiment, you will check whether radiation-generated GM counts are consistent with Poisson statistics. You will do it by repeating measurements many times, and examining the mean values of the cumulants calculated from these measurements, as described in Section 4.
We want sufficient accuracy to check if $K_m^{(i)}$ with $i = 1, 2, 3$ are close enough to $\kappa^{(3)} = \lambda$, e.g. $\mathrm{STD}(K_m^{(3)}) < \lambda/3$. For this we need quite a large number of measurements: $m > 9 \cdot 15\lambda$, see Appendix A.2. A similar criterion for $K_m^{(2)}$ is less demanding. In order to keep the overall measurement time reasonable, we want to chose a short time for a single measurement and a small count rate so $\bar{n} \to \lambda$ is small, and therefore, m is not very large.

1. Take a relatively weak source, for instance Cesium, and put it with label up to reduce the radiation flux.

2. Set the measurement time to, for example, 10 s and calculate the average rate. You want it to be, say, $\bar{n} \approx 5$ counts per second to keep m reasonably small. If the rate is too high – increase the distance to the tube. If it is too small – maybe turn the source upside down or pick another source.

3. Take a set of at least $m = 150\bar{n}$ 1 second measurements. For $\bar{n} = 5$ it will take some 12 minutes–not that long.

4. Calculate $K_m^{(1)} = \bar{n}$ and $K_m^{(2)} = \mathrm{VAR}(n) = \sum_i (n_i - \bar{n})^2/(m-1)$ for the new large set of data.

5. Calculate $K_m^{(3)} = \mathrm{VAR}(n) = \sum_i (n_i - \bar{n})^3/(m-1)$. Compare it with $\bar{n}$. For Poisson process we expect $\bar{n}$, $K_m^{(2)}, K_m^{(3)} \to \lambda$ for $m \to \infty$. Compare it also with the expectation for the Gaussian statistic: $\kappa^{(3)} = 0$.

6. Plot the probability $P(n)$ to get n counts as obtained from the measurements, and the Poisson and Gaussian distributions with the same $\bar{n}$.

6.3 Attenuation of β-radiation and estimation of maximal β-particle energy

Objective: This experiment aims at investigation attenuation of β-radiation by varying the thickness of aluminum plates between the source and detector. The obtained range in Al will be used to find the maximal β-particle energy for a few sources. You will use the Tenth-Value Layer (TVL) criterium to determine the range.

1. Place a ^{204}Tl radioactive source on the third shelf from the top and begin taking data.
2. Measure the detection rate with accuracy $\approx 3\%$.
3. Place an aluminum piece on the top shelf and take another data run.
4. Depending on the degree of the radiation attenuation by the piece, decide what the next piece do you want to place.
5. Repeat this a minimum of 7 more times. Each time add the data point to the plot of the counting rate vs. aluminum thickness. You want to have enough points to find the range. You can use the upper shelves to combine the pieces and get the sum of their thicknesses. You should increase thickness until the count rate drops at least ten times; if time permits, until it is close to the background level. Al low count rates you might need to sacrifice accuracy to save time.
6. Plot the count rate vs aluminum thickness both is linear and logarithmic scale. Find the β-particle range.
7. Repeat the procedure with a ^{90}Sr and compare the results.

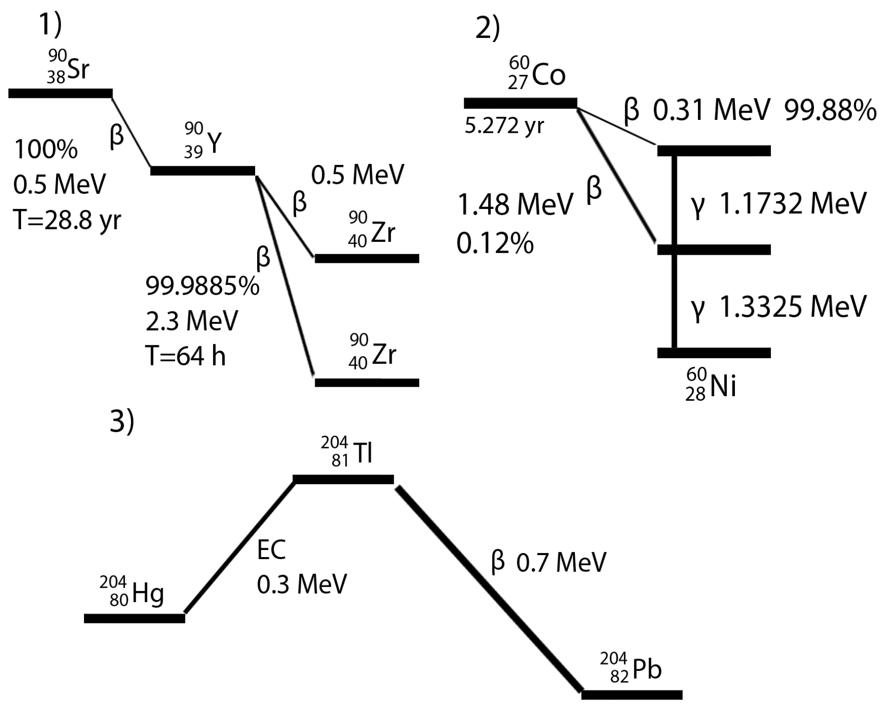

Figure 6: Decay scheme for (1) Strontium, (2) Cobalt, and (3) Thallium

8. Do the same for ^{60}Co. Pay attention on the decay scheme. Why the results are different?

6.4 Alpha particles range

Objective: This experiment aims at measuring the α-particle range and finding their energy by systematically varying the distance between the GM detector and the α-particle source. In this case, the attenuation media is air. We are interested in the attenuation of the particle flux per unit spherical angle, not just the counting rate. Since α-particle range is just a few centimeters, is comparable with the GM window diameter, and will be changed in the experiment, the spherical angle Ω at which the detector is seen from the source will change significantly. Therefore, you will need to normalize the rate by the Ω, Eq.13. You will use ^{210}Po as an α-particle source. Important: place the source with the label down, otherwise it would block alpha particle propagation.

1. Place the radioactive source on the 1st shelf and measure the count rate. Use the same considerations to balance the accuracy and time.

2. Since the range of the α-particles in air is just a few centimeters, use spacers to bring the source closer to the detector. Take a few measurements with the source approaching the top of the box.

3. Move the source down one shelf and take the data again. If the rate is too small, say less than 1 cps, put spacers over the second shelf to get a reasonable count rate. *Plot the data as you measure to determine the distances at which to collect more data.*

4. Subtract the background and calculate the source-to-GM distance using taking into account 12.3 mm distance from the top shelf to the detector window. In order to obtain radiation flux per unit area, divide the count rate by the solid angle at which the tube sees the source, Eq.13.

5. Plot the result as a function of the distance *from the GM window.* Find the range using TLV criterium.

6. Take into account the areal density of the GM window $2mg/cm^2$. For this translate the areal density to the respective length of the air, assuming that the range *in distance* is inversely proportional to the media density. Take air density to be $1.3\,\mathrm{mg/cm^2}$.

7. Find the α-particle energy from the range you found using Eq. 5

A Appendices

A.1 Poisson distribution

The Poisson distribution can be obtained from the binomial one for N independent tries. The probability to detect n events out of N attempts is:

$$P_N(n) = \binom{N}{n} p^n (1-p)^{N-n}, \tag{14}$$

where p is the probability to observe the event in a single attempt. We now take N to infinity and p to zero, while keeping the expectation value of the success number $\lambda = p \cdot N$ constant. We now observe that:

$$\lim_{N\to\infty} p^n \binom{N}{n} = \lim_{N\to\infty} \frac{(pN)^n}{n!} = \frac{\lambda^n}{n!}$$

$$\lim_{N\to\infty}(1-p)^{N-n} = \lim_{N\to\infty}(1-\lambda/N)^{N-n} = \lim_{N\to\infty}(1-\lambda/N)^{N} = e^{-\lambda}$$

We end up with the Poisson distribution of the probability to observe n events:

$$P(n) = \frac{\lambda^n}{n!}e^{-\lambda} \tag{15}$$

A.2 Number of measurements required to estimate $\kappa^{(i)}$

Below we estimate the number of measurements required to confidently distinguish between the distributions on the basis of their cumulants $\kappa^{(i)}$. We shall restrict ourselves with the first 3 cumulants, that coincide with the central moments.

Let us start with recalling the following standard result: Consider an arbitrary variable x with average $\langle x\rangle = 0$ and variance $\langle x^2\rangle = \sigma_x^2$. Suppose we do m *independent* measurements of x and the mean: $y_m = \overline{x}_m = (1/m)\sum_{i=1}^m x_i$. Then, $\sigma_{y_m}^2 = \langle y_m^2\rangle = \sigma_x^2/m$, or $\sigma_{y_m} = \sigma_x/\sqrt{m}$. Now we apply this to uncertainty of the cumulants.

1. The first cumulant: $\kappa^{(1)} = \langle n\rangle = \lambda$. The variance of the measured $K_m^{(1)}$ will be expressed through the second cumulant $\kappa^{(2)} = \langle (n-\lambda)^2\rangle = \lambda$. We define $x = n-\lambda$, and y as above.
$$\langle x^2\rangle = \langle (n-\lambda)^2\rangle = \lambda$$
$$\sigma_{K_m^{(1)}} = \sigma_{y_m} = \sqrt{\lambda/m}, \text{ and } \sigma_{K_m^{(1)}}/\kappa^{(1)} = 1/\sqrt{\lambda m} \tag{16}$$

2. The second cumulant: $x = (n-\lambda)^2 - \lambda$; $\langle x\rangle = 0$
$$\langle x^2\rangle = \langle((n-\lambda)^2-\lambda)^2\rangle = \langle(n-\lambda)^4\rangle - 2\lambda\langle(n-\lambda)^2\rangle + \lambda^2 = \langle(n-\lambda)^4\rangle - \lambda^2$$
The 4th central moment $\langle(n-\lambda)^4\rangle$ can be directly obtained from the Poisson distribution. However, for an estimate in the leading order in $\lambda \gg 1$ it is enough to use the Central Limit Theorem, and approximate $\langle(n-\lambda)^4\rangle$ by the Gaussian result A.3.
Anyway, $\langle(n-\lambda)^4\rangle \approx 3\lambda^2$, and therefore:
$$\sigma^2_{K_m^{(2)}} = 2\lambda^2/m, \text{ and } \sigma_{K_m^{(2)}}/\kappa^{(2)} = \sqrt{2/m} \tag{17}$$

3. The third cumulant: $x = (n-\lambda)^3 - \lambda$; $\langle x\rangle = 0$
$$\langle x^2\rangle = \langle((n-\lambda)^3-\lambda)^2\rangle = \langle(n-\lambda)^6\rangle - 2\lambda\langle(n-\lambda)^3\rangle + \lambda^2 = \langle(n-\lambda)^6\rangle - \lambda^2$$
Similarly to the above, $\langle(n-\lambda)^6\rangle \approx 15\lambda^3$. and therefore:
$$\sigma^2_{K_m^{(3)}} = 15\lambda^3/m, \text{ and } \sigma_{K_m^{(3)}}/\kappa^{(3)} = \sqrt{15\lambda/m} \tag{18}$$

A.3 Central moments of Gaussian distribution

For a Gaussian distributed value x with $\langle(x-\langle x\rangle)^2\rangle = \sigma^2$ all central moments are: $\langle(x-\langle x\rangle)^{2n}\rangle = (2n-1)!!\sigma^{2n}$.

A.4 Sources

Radiation units: 1 curie – $3.7{\cdot}10^{10}$ decays per second (dps). 1dps=1 becquerel (Bq).
Thallium-204 $^{104}_{81}$Tl β-decays into $^{104}_{80}$Pb with $E_{max} = 0.7\,$MeV.
Strontium-90 ($^{90}_{38}$Sr) β-decays into $^{90}_{39}Y$ with $E_{max} = 0.546\,$MeV. The Yttrium isotope's half-life time is 28.8y, and it in turn undergoes β-decay with half-life time of 64 hours and $E_{max} = 2.28\,$MeV into $^{90}_{40}$Zr (zirconium), which is stable. Note that Sr/Y is almost a pure beta particle source; the gamma photon emission from the decay of ^{90}Y is so infrequent that it can normally be ignored.
Cobalt-60 ($^{60}_{27}Sr$) is mostly a γ-source with energies 1.17 and 1.33 MeV.

References

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